ETH zürich ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads

Integrated Systems Laboratory (ETH Zürich)

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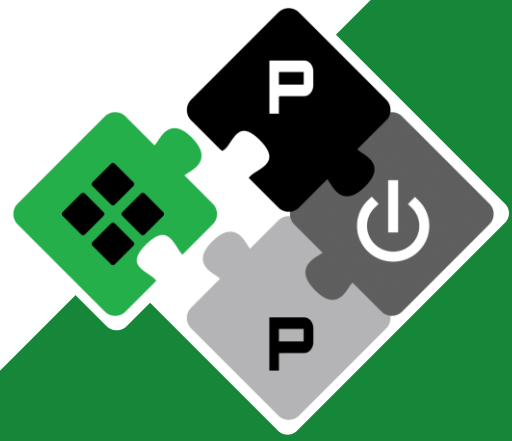

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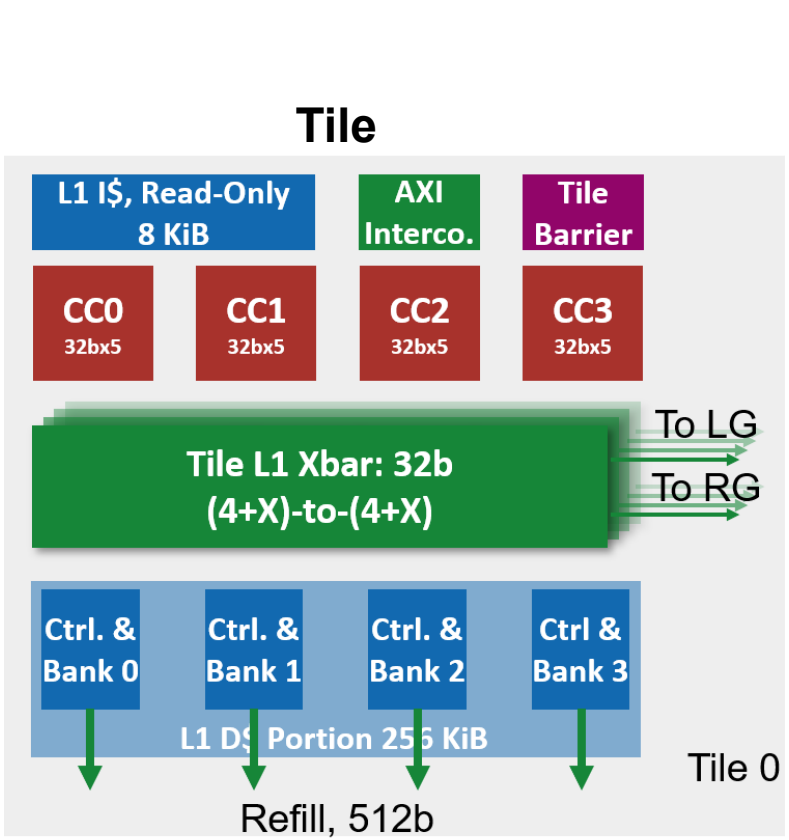
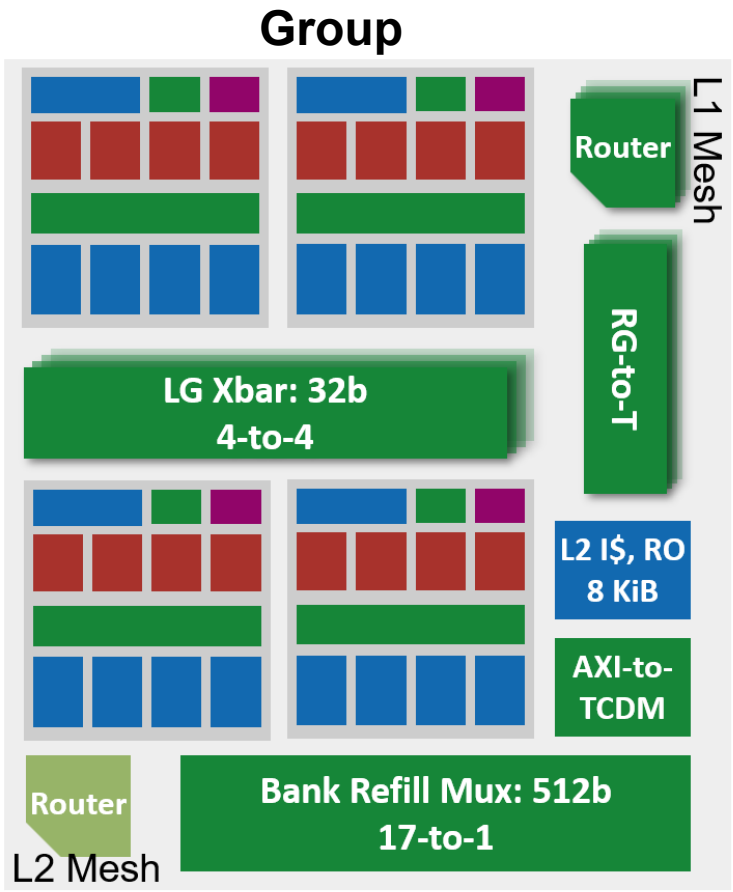
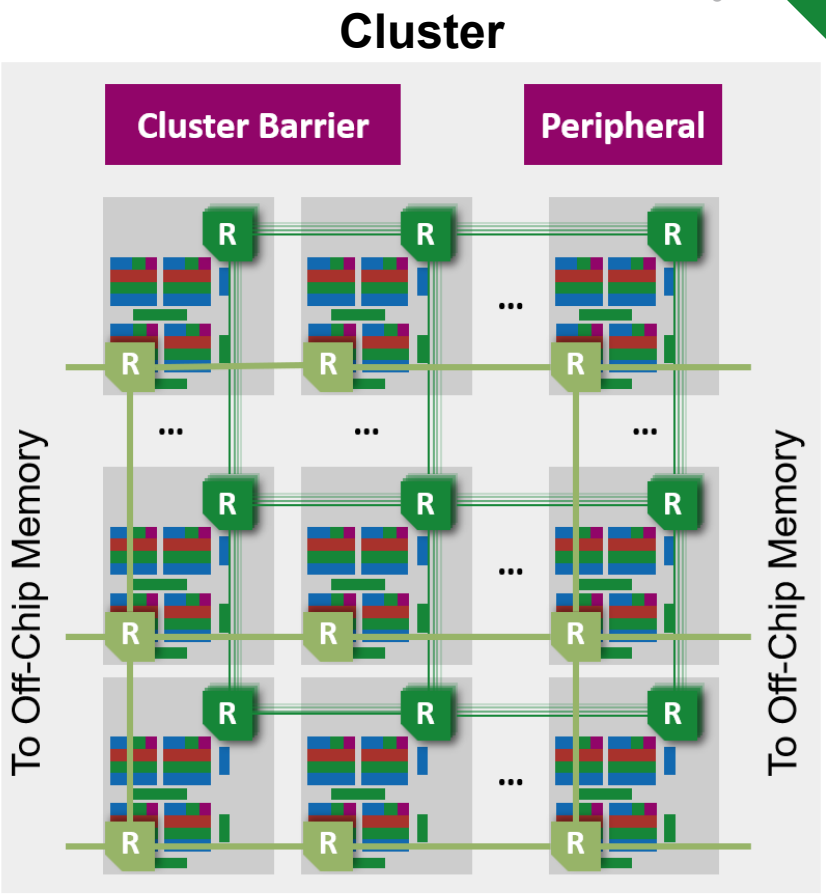
Hardware Development (Cluster)







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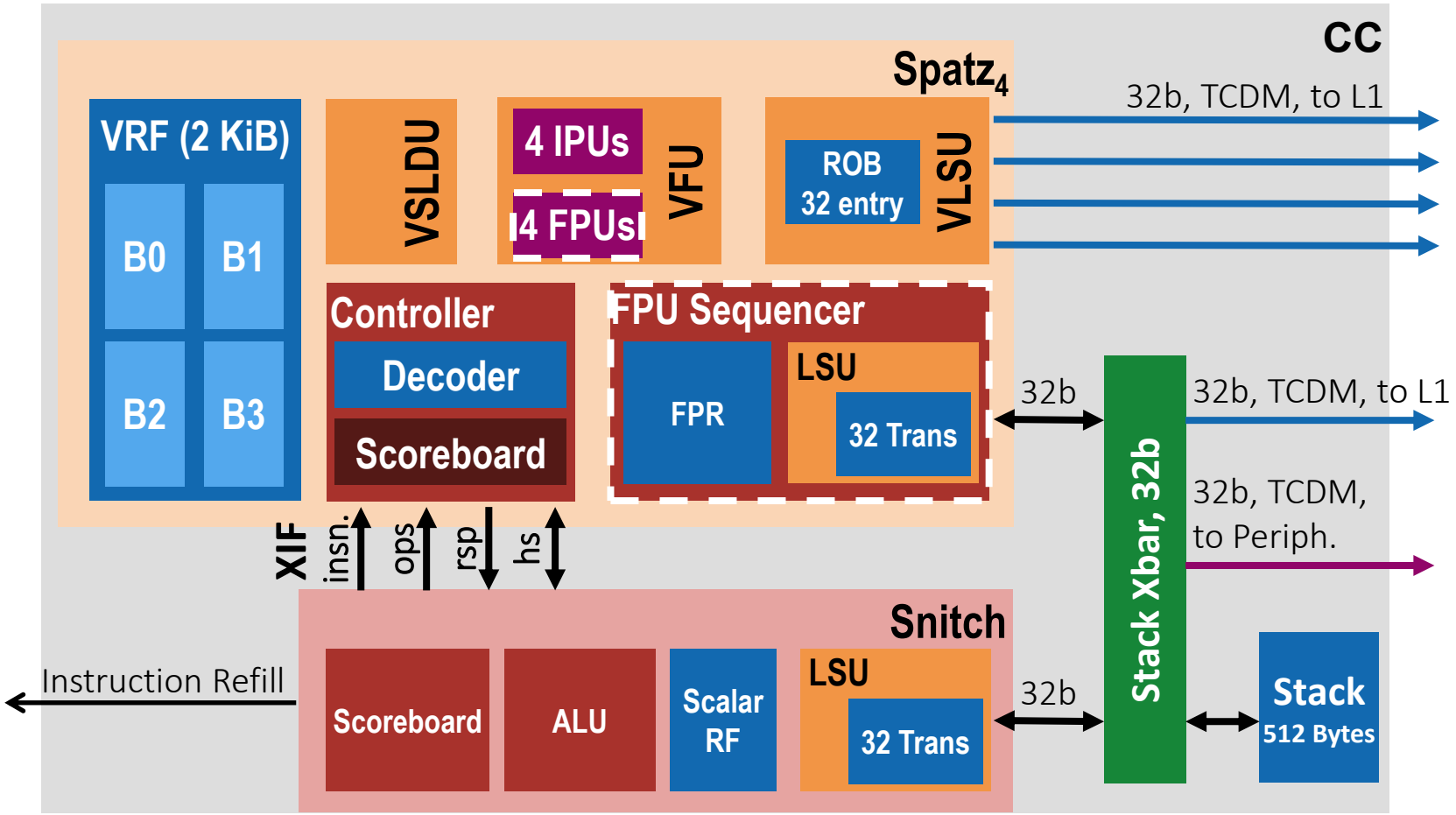

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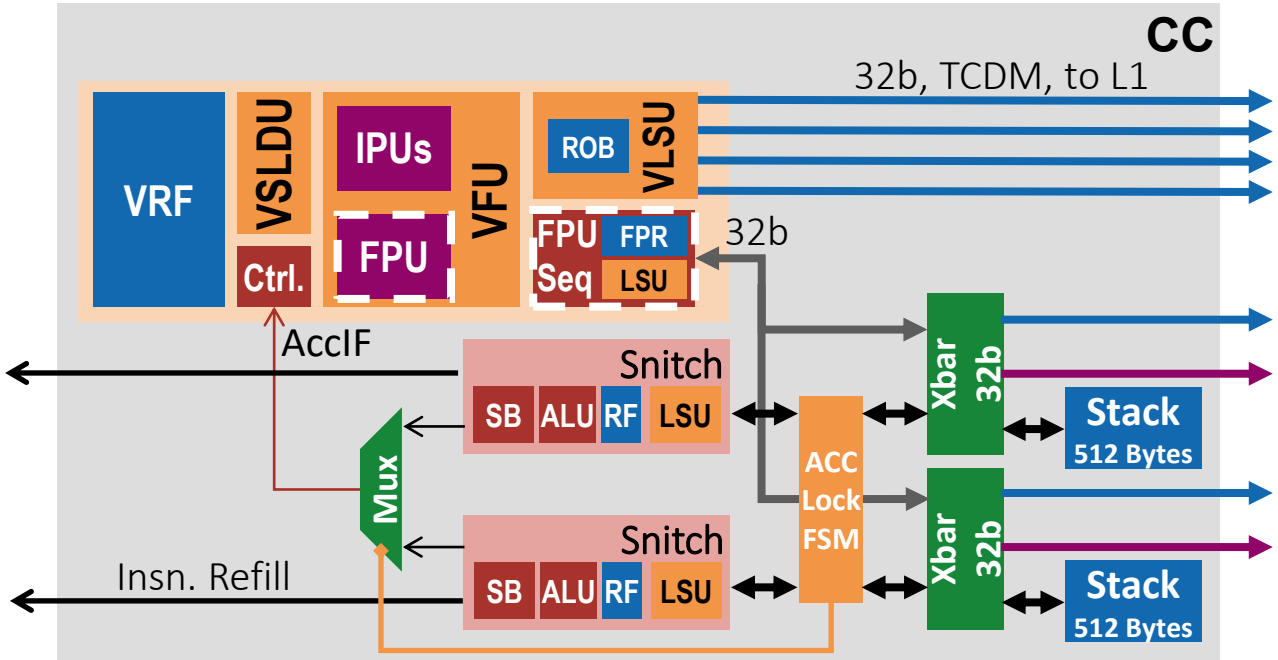
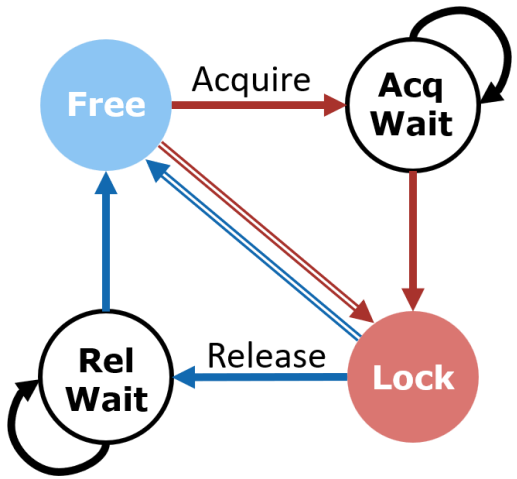


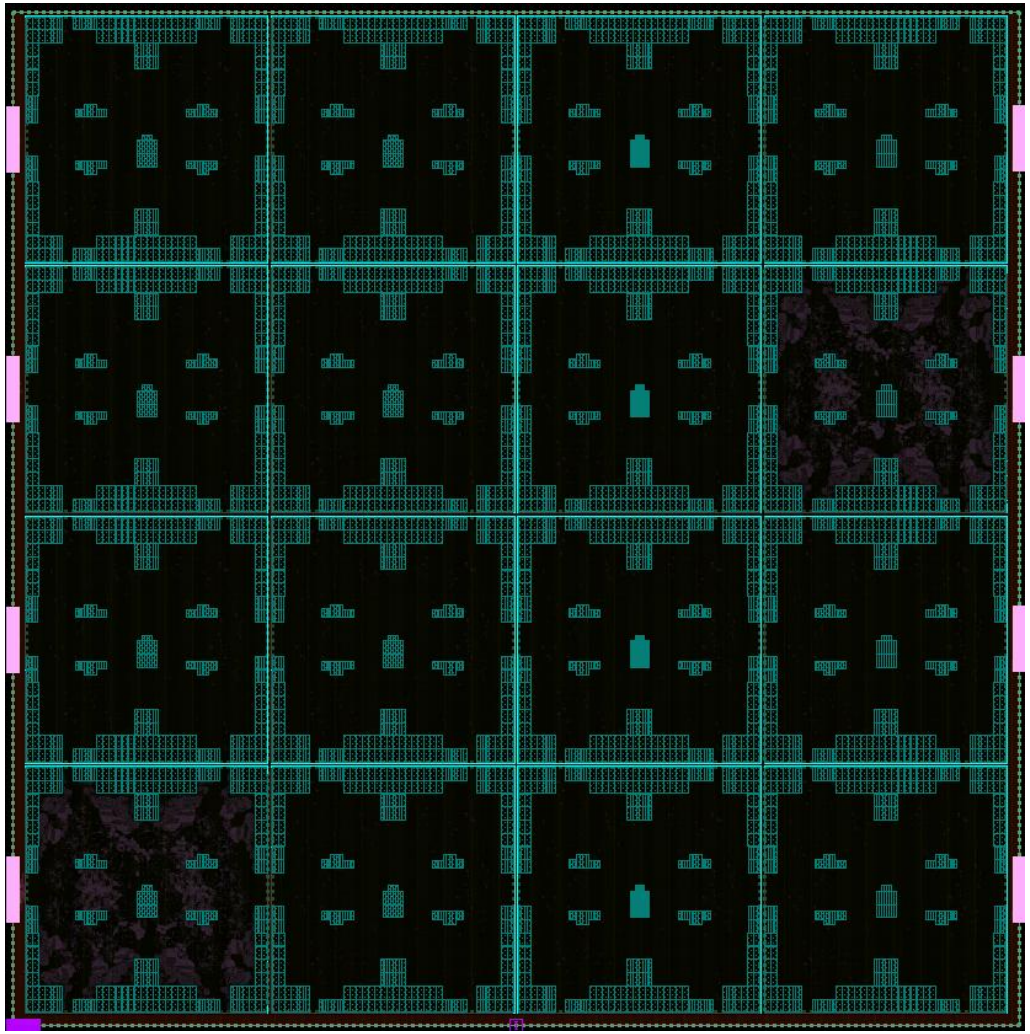


- **Free mode**
  - Round-robin requests from both Snitch to Spatz
  - Default mode on booting, so that both Snitch can clear FPR and VRF if using uniformed bootrom.
- **Lock mode**
  - Any Snitch can lock the access by writing to a mem-mapped address.
  - The write will be taken by the FSM to switch mode
  - Pause acc access during switch, and switch completes when no running acc insn.
  - Spatz “private” to the owner

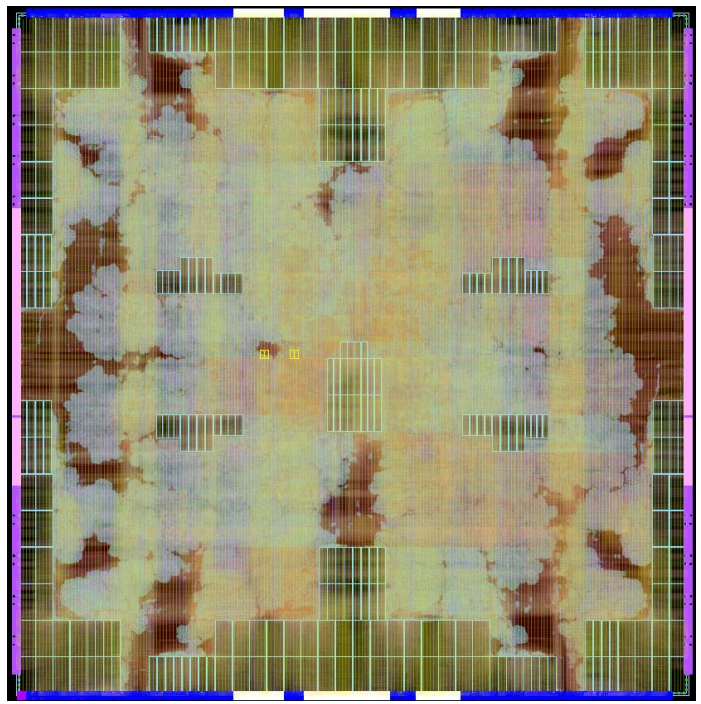
Hardware Development (Cluster)

- **Developments**
  - Rebase the multi-group configuration with the folded cache
    - Development branch is now merged into main
    - Functionally correct, performance checking and debugging undergoing since RTL simulation runs quite slow
  - Add dual-scalar core support
    - Currently tested on 4-group configuration for faster simulation speed
    - PR opened, still needs some cleaning and testing jobs

Physical Design



- **Started a physical design run on the latest main (folded + multi-group)**
  - Timing: same 200ps path in cache controller (cluster-level P&R undergoing)
  - Routing: 17 shorts => Routing clean

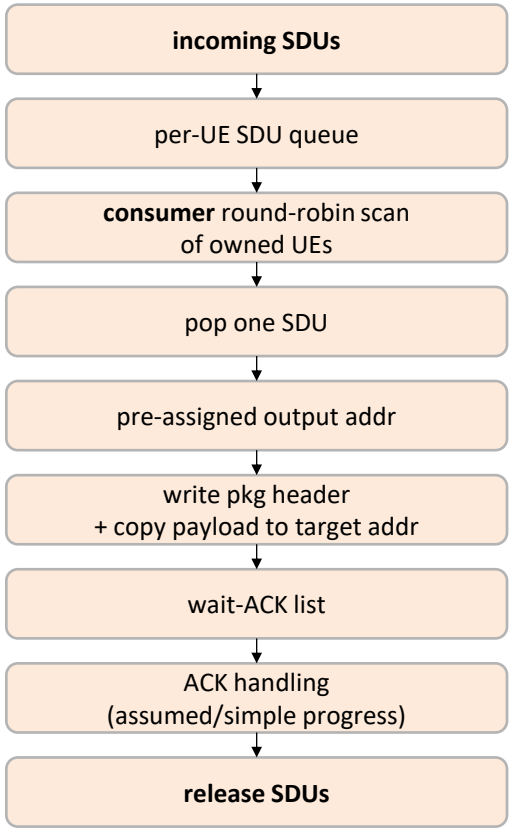


HW dev: LR/SC bug fix

- **Reservation overwrite**
  - LR from another core could overwrite a live reservation → starvation.
  - **Fix:** keep current reservation + ResvTimeoutCycles (1024 cyc)
- **Incorrect SC result**
  - Single outstanding-SC register could misreport a zero memory word as SC success
  - **Fix:** carry is_sc/sc_fail in tcdm_user_t; avoid single-entry SC tracking.
- **Validation**
  - New test: lrsc-forward-progress
    - 16 cores, symmetric CAS-increment retry loop on one shared word
    - **Before:** 22/32 completed; 8 iter/core stuck
    - **After:** 256/256 completed; 16 iter/core in 25,778 cycles (~100 cycle per lr/sc pair)

SW dev: RLC Plan Stage + Processing Stage

Legacy kernel (direct copy path)



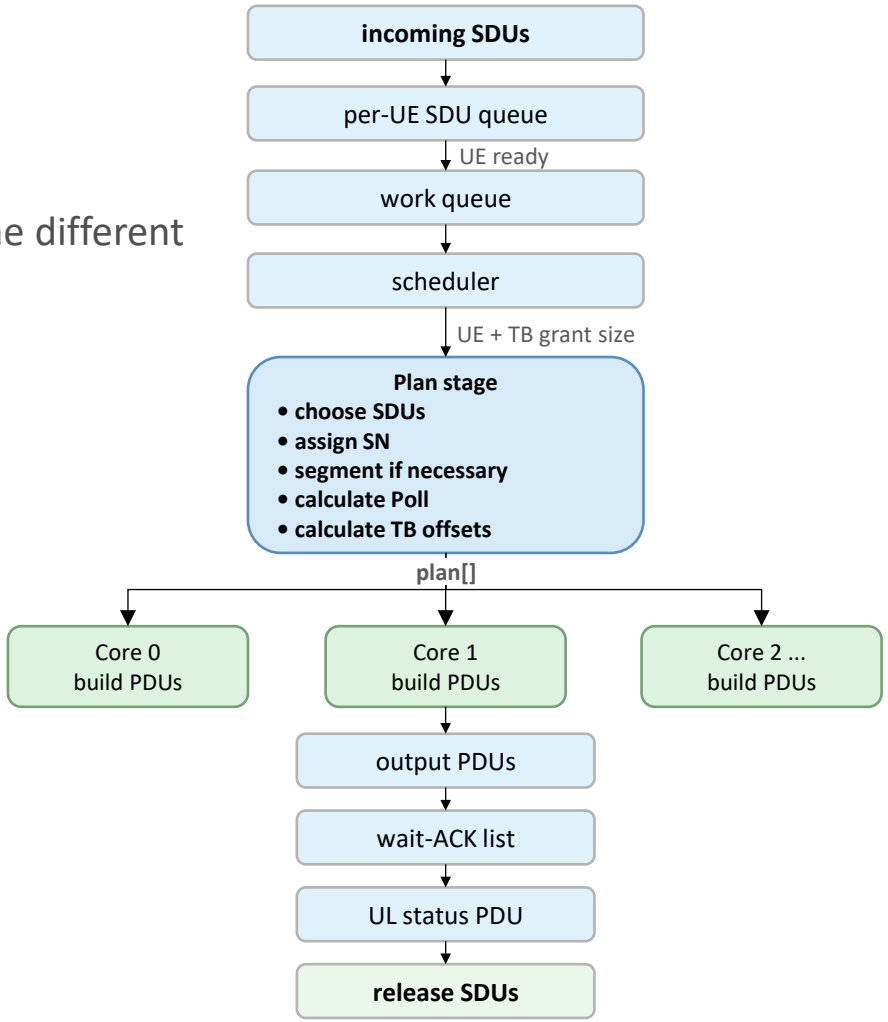
Limitation:

- No resource allocation algorithm involved;
- Static task scheduling: fixed pkg producers and consumers
- No TTI control

New AM kernel (plan + parallel execute)

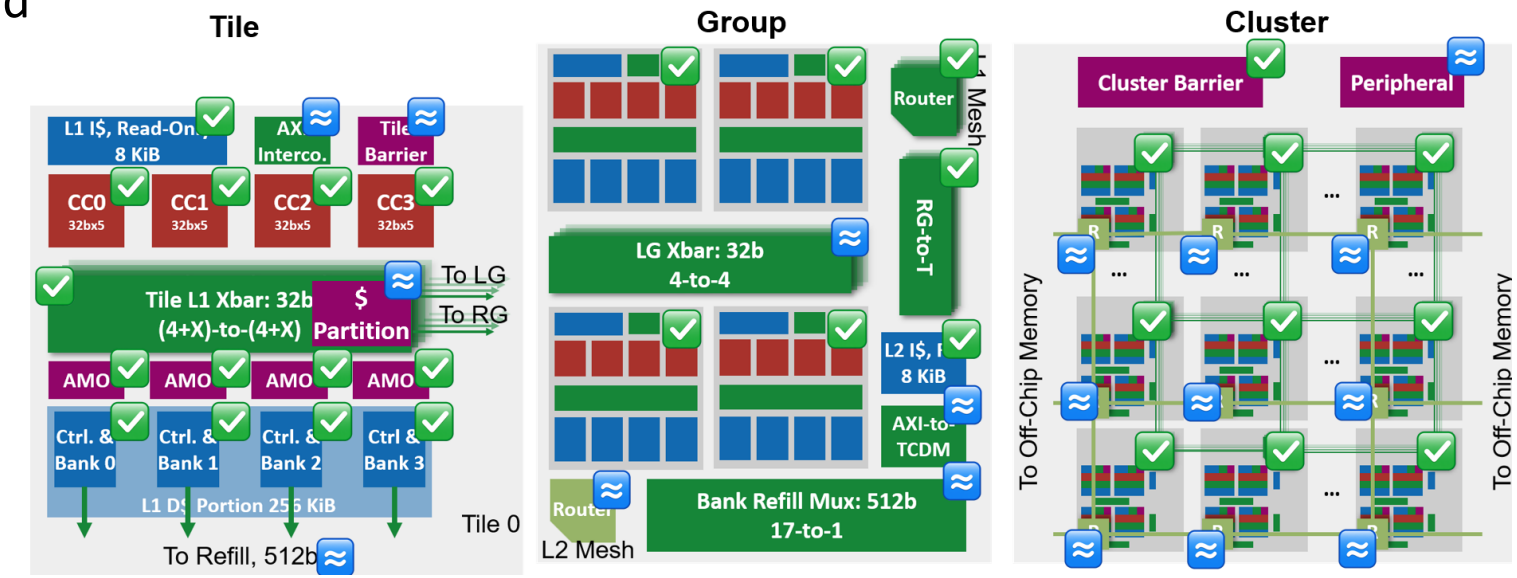
Main idea: Separate the different tasks:

- Pkg receiving task
- Plan task
- Pkg assembly task
- UE status task



GVSoC Modeling Progress

- **L2 Refill Mesh Fixed**
  - Topology now matches RTL: 6×4 mesh, 16 routers, 8 channels
  - Fixed 1 KB channel interleave and 512 MiB DRAM
- **Calibration Improved**
  - Better NoC latency modelling
  - 64-core perf gap reduced from +60.8% → +14.8%
    - Used old RLC kernel
    - 2x2 groups x 4 tiles x 4 cores



Next Step

- **ICCD 2026**
  - Unfortunately, our submission to ICCD was rejected
    - Seems like reviewer are not in this field at all…
      From one reviewer: *I'm not sure its novelty, but its novelty could be a key factor in the paper's acceptance.*
    - Waiting for Luca's inputs to retarget, most likely DATE 2027
    - Also wait for his final approval to put it on axiv
- **Diyou Side**
  - Performance evaluation and debugging
  - Paper adjustment and resubmission
- **Zexin Side**
  - GVSoC development and calibration under multi-group config, add dual-scalar core support
  - SW development based on the benchmark plan, adapt features from Johannes's kernel

Timeline

		2025									2026			
		11	12	1	2	3	4	5	6	7	8	9	10	11
WP1 ManyCore Architecture	Multi-Tile Scaling [RTL] Complete													
	4/16-Group Scaling [GVSoC + RTL] Complete													
	Interco. Optimization (multi-group) [RTL] Almost Complete													
	Timing optimization for NoC [RTL] Ongoing by colleague													
	NoC Topology Exploration [RTL, GVSoC] Ongoing by student													
WP2 Cache	Cache partitioning [RTL] Complete													
	WT-L1 by master student [RTL] Complete													
	Folded Cache [GVSoC + RTL] Complete													

Timeline

		2025										2026		
		11	12	1	2	3	4	5	6	7	8	9	10	11
WP3 VPE	Shared vector core [RTL + GVSoC] Almost Complete													
WP4 Benchmark	Single-user kernel optimization [SW] Ongoing													
	Multi-user and more scenarios [SW] Ongoing													
Report	Final Report and Benchmark													

L1 Access Latency

- **Problem: increased L1 access latency when scaling up**
- **Solutions**
  - **(Complete, RTL & Physical) Hardware optimization**
    Further optimize the interconnection and cache controller to reduce access latency on existing architecture
    => Retiming the current interconnection, remove unnecessary cuts in interco and cache
    => Target to reduce 1-2 cycles (~5 cyc) for hit-to-use latency
    Currently has reduced to 6 cycles:
    Core => Xbar => AMO => Ctrl (2) => AMO => Core
  - **(Complete, RTL) Cache partitioning**
    Partition the shared L1 to keep data localized to avoid remote access latency
  - **(Complete with some problems, RTL & Physical) WT-L1**
    Explore and evaluate a private WT L1 for each core
    => Performance degrade on single tile
    => Needs more work to support larger system

L1 Access Latency

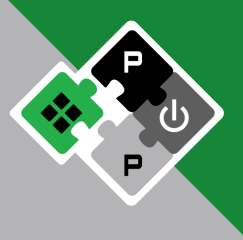
- **Problem: large NoC latency (~28 cyc based on previous exploration)**
- **Solutions**
  - **(Ongoing, Physical) Reduce hop latency**
    Explore the timing under 7nm technology to see the possibility of reducing per-hop latency to 1 cycle
    => Reduce the latency by 1/2
  - **(Planned, Physical) Cut clk-domain (optional)**
    If 1-cyc is not possible, explore to operate the NoC under a higher frequency than cores (e.g. 1.5GHz)
    => Effectively cut the latency by 1/3
  - **(Planned, GVSoC & Physical) NoC Topology**
    Explore different NoC topologies
    => Torus to reduce the diameter of NoC by 1/2, reduce longest latency by 1/2

PE-Related

- **Problem: Improve scalar core performance**
- **Solutions**
  - **(Complete, RTL & GVSoC) Shared vector core**
    Explore two Snitch sharing one Spatz configuration to improve scalar performance
    Software complexity needs to be considered, especially on the possible conflicting use of register file.
    Discussed Solution: critical section

RLC Workload

- **Problem: Current RLC model cannot reveal all properties of the kernel**
- **Solutions**
  - **(Ongoing, SW) Kernel optimization**
    Adapt the current model to better represent the real cases, e.g. do some operations on the data. Further cooperation with HQ side to develop a more complete and better model.
  - **(Planned, SW) Kernel development**
    Implement the missing use cases for multi-user



Thank you!